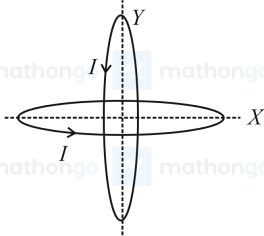


- A long straight wire of circular cross-section (radius a) is carrying steady current I . The current I is uniformly distributed across this cross-section. The magnetic field is
[2023 (06 Apr Shift 1)]
(1) inversely proportional to r in the region $r < a$ and uniform throughout in the region $r > a$
(2) directly proportional to r in the region $r < a$ and inversely proportional to r in the region $r > a$
(3) Zero in the region $r < a$ and inversely proportional to r in the region $r > a$
(4) uniform in the region $r < a$ and inversely proportional to distance r from the axis, in the region $r > a$

- Two identical circular wires of radius 20 cm and carrying current $\sqrt{2}$ A are placed in perpendicular planes as shown in figure. The net magnetic field at the centre of the circular wires is $\times 10^{-8}$ T.



(Take $\pi = 3.14$)

[2023 (06 Apr Shift 1)]

- A proton with a kinetic energy of 2.0 eV moves into a region of uniform magnetic field of magnitude $\frac{\pi}{2} \times 10^{-3}$ T. The angle between the direction of magnetic field and velocity of proton is 60° . The pitch of the helical path taken by the proton is ____ cm. (Take, mass of proton = 1.6×10^{-27} kg and charge on proton = 1.6×10^{-19} C).

[2023 (06 Apr Shift 2)]

- A charge particle moving in magnetic field B , has the components of velocity along B as well as perpendicular to B . The path of the charge particle will be
[2023 (08 Apr Shift 1)]

- (1) helical path with the axis perpendicular to the direction of magnetic field B
- (2) helical path with the axis along magnetic field B
- (3) circular path
- (4) straight along the direction of magnetic field B

- The magnetic intensity at the centre of a long current carrying solenoid is found to be 1.6×10^3 A m^{-1} . If the number of turns is 8 per cm, then the current flowing through the solenoid is ____ A.

[2023 (08 Apr Shift 1)]

- The ratio of magnetic field at the centre of a current carrying coil of radius r to the magnetic field at distance r from the centre of coil on its axis is $\sqrt{x} : 1$. The value of x is ____.

[2023 (08 Apr Shift 2)]

- Given below are two statements:

Statement I: If the number of turns in the coil of a moving coil galvanometer is doubled then the current sensitivity becomes double.

Statement II: Increasing current sensitivity of a moving coil galvanometer by only increasing the number of turns in the coil will also increase its voltage sensitivity in the same ratio

In the light of the above statements, choose the correct answer from the options given below:

[2023 (10 Apr Shift 1)]

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are false
- (4) Both Statement I and Statement II are true

- A straight wire carrying a current of 14 A is bent into a semicircular arc of radius 2.2 cm as shown in the figure. The magnetic field produced by the current at the centre O of the arc is $\times 10^{-4}$ T



[2023 (10 Apr Shift 2)]

9. An electron is allowed to move with constant velocity along the axis of current carrying straight solenoid.

- (A) The electron will experience magnetic force along the axis of the solenoid.
- (B) The electron will not experience magnetic force.
- (C) The electron will continue to move along the axis of the solenoid.
- (D) The electron will be accelerated along the axis of the solenoid.
- (E) The electron will follow parabolic path-inside the solenoid.

Choose the correct answer from the option given below:

[2023 (11 Apr Shift 2)]

- (1) B, C and D only
- (2) A and D only
- (3) B and C only
- (4) B and E only

10. An electron is moving along the positive x-axis. If uniform magnetic field is applied parallel to the negative z-axis, then

- A. The electron will experience magnetic force along positive y-axis
- B. The electron will experience magnetic force along negative y-axis
- C. The electron will not experience any force in magnetic field
- D. The electron will continue to move along the positive x-axis
- E. The electron will move along circular path in magnetic field

Choose the correct answer from the option given below:

[2023 (13 Apr Shift 2)]

- (1) A and E only
- (2) C and D only
- (3) B and E only
- (4) B and D only

11. A straight wire AB of mass 40 g and length 50 cm is suspended by a pair of flexible leads in uniform magnetic field of magnitude 0.40 T as shown in the figure. The magnitude of the current required in the wire to remove the tension in the supporting leads is _____ A. (Take $g = 10 \text{ m s}^{-2}$).



[2023 (13 Apr Shift 2)]

12. An electron in a hydrogen atom revolves around its nucleus with a speed of $6.76 \times 10^6 \text{ m s}^{-1}$ in an orbit of radius $0.52 \text{ \AA}$. The magnetic field produced at the nucleus of the hydrogen atom is _____ T.

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (2) 2. (628) 3. (40) 4. (2) 5. (2) 6. (8) 7. (1) 8. (2)
9. (3) 10. (3) 11. (2) 12. (40)

1. (2)

The formula to calculate the magnetic field inside the conducting wire is given by

$$B_{\text{in}} = \frac{\mu_0 J r}{2} \dots (1)$$

The formula to calculate the magnetic field outside the conducting wire is given by

$$B_{\text{out}} = \frac{\mu_0 J a^2}{2r} \dots (2)$$

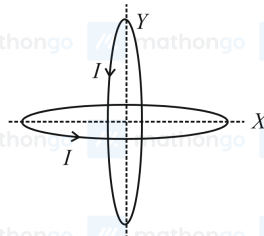
From equation (1), it can be concluded that the magnetic field is directly proportional to the radial distance inside the wire, whereas from equation (2), it can be concluded that the magnetic field is inversely proportional to the radial distance outside the wire.

2. (628)

The formula to calculate the magnetic field (B) at the centre of a current carrying circular loop is given by

$$B = \frac{\mu_0 i}{2R} \dots (1)$$

where, μ_0 is the permeability of free space, i is the current and R is the radius of the loop.



B_V , B_H are the magnetic field due to the vertical and the horizontal loop, which by magnitude are equal.

The resultant magnetic field (B_{net}) at the centre is, then, given by

$$\begin{aligned} B_{\text{net}} &= \sqrt{B^2 + B^2} = \sqrt{2}B \\ &= \frac{\sqrt{2} \times 4\pi \times 10^{-7} \times \sqrt{2}}{2 \times 0.2} \\ &= 628 \times 10^{-8} \text{ T} \end{aligned}$$

3. (40)

The given data is

$$K.E = 2 \text{ eV}$$

$$B = \frac{\pi}{2} \times 10^{-3} \text{ T}$$

$$\theta = 60^\circ$$

The pitch of the proton is given by

$$P = \frac{2\pi m}{qB} \times v \cos \theta \dots (i)$$

By using

$$\begin{aligned} K &= \frac{mv^2}{2} \\ \Rightarrow v &= \sqrt{\frac{2K}{m}} \end{aligned}$$

Substituting the values in equation (i)

$$\begin{aligned} &= \frac{2\pi \times \sqrt{2mKE} \times \frac{1}{2} \times 2}{1.6 \times 10^{-19} \times \pi \times 10^{-3}} \\ &= \frac{2 \times \sqrt{2 \times 1.6 \times 10^{-27} \times 2 \times 1.6 \times 10^{-19}} \times 10^3}{1.6 \times 10^{-19}} \\ &= 2 \times 2 \times 10^{-1} = 0.4 \text{ m} \end{aligned}$$

4. (2)

The formula to calculate the radius of the path followed by the charged particle under an external magnetic field can be written as

$$r = \frac{mv}{Bq} \dots (1)$$

The perpendicular component of velocity results in circular motion, while the parallel component results in linear motion. Hence, the path of the charged particle is helical in nature, the axis of which lies along the direction of the magnetic field.

5. (2)

The magnetic intensity H is defined as the product of the number of turns per unit length in a coil n and the current that it carries I .

The magnetic field of a solenoid is given by $B = \mu_0 ni$

$$H = \frac{B}{\mu_0} = \frac{\mu_0 ni}{\mu_0}$$

$$\Rightarrow H = ni \dots (i)$$

The given data is

$$H = 1.6 \times 10^3 \text{ A m}^{-1}$$

$$n = 8 \text{ per cm}$$

Substituting the values in equation (i)

$$\Rightarrow 1.6 \times 1000 = \frac{8}{100} \times i$$

$$\Rightarrow i = 2 \text{ A}$$

6. (8)

The formula to calculate the magnetic field at any axial point situated at a distance x from the centre of a circular current carrying conductor of radius r can be written as

$$B = \frac{\mu_0 I r^2}{2(r^2 + x^2)^{3/2}} \dots (1)$$

Substitute 0 for x in equation (1) to calculate the magnetic field (B_c) at the centre of the conductor.

$$B_c = \frac{\mu_0 I r^2}{2(r^2 + 0^2)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2r^3}$$

$$= \frac{\mu_0 I}{2r} \dots (2)$$

Substitute r for x in equation (1) to calculate the magnetic field (B_r) at the given distance from the centre of the conductor.

$$B_r = \frac{\mu_0 I r^2}{2(r^2 + r^2)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{4\sqrt{2}r^3}$$

$$= \frac{\mu_0 I}{4\sqrt{2}r} \dots (3)$$

Divide equation (2) by equation (3) to calculate the required ratio of the magnetic fields.

$$\frac{B_c}{B_r} = \frac{\frac{\mu_0 I}{2r}}{\frac{\mu_0 I}{4\sqrt{2}r}}$$

$$= 2\sqrt{2}$$

$$= \sqrt{8}$$

7. (1)

For a moving coil galvanometer,

$$(NiAB) = k\theta$$

The current sensitivity is given by

$$S_i = \frac{NAB}{k}$$

So, if N is doubled then, the current sensitivity is also doubled.

The voltage is given by

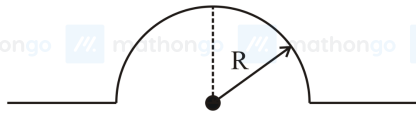
$$V = \left(\frac{BNA}{Rk} \right) \theta$$

The voltage sensitivity is given by

$$S_v = \left(\frac{NAB}{kR} \right)$$

As, N is increased, R also increases. So, the voltage sensitivity does not change as $\frac{N}{R}$ remains constant.

8. (2)



It is given that

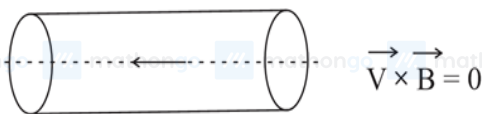
$$i = 14 \text{ A}$$

$$R = 2.2 \times 10^{-2} \text{ m}$$

The magnetic field at the centre O is

$$\begin{aligned} B &= \frac{\mu_0 i}{4R} \\ &= \frac{4\pi \times 10^{-7} \times 14}{4 \times 2.2 \times 10^{-2}} \\ &= 2 \times 10^{-4} \text{ T} \end{aligned}$$

9. (3)



$$\vec{F} = 0$$

The Lorentz force is given by $\vec{F} = q(\vec{v} \times \vec{B}) = qvB \sin \theta \hat{n}$. The angle $\theta = 0^\circ$. Hence the force experienced is zero.

Thus, the electron will not experience any magnetic force and the electron will continue to move along the axis.

10. (3)

The Lorentz force is given by

$$\begin{aligned} \vec{F} &= q\vec{v} \times \vec{B} \\ &= -e\vec{v} \times \vec{B} \end{aligned}$$

$$\text{So, } \vec{F} = \hat{i} \times \hat{k} = -\hat{j}.$$

$\vec{F}$ is along $-y$ axis

As the magnetic field is perpendicular to the velocity, motion would be circular.

11. (2)

The formula to calculate the magnetic force on the conductor is given by

$$F_B = ILB \dots (1)$$

The formula to calculate the gravitational force on the conductor is given by

$$F_G = mg \dots (2)$$

Equate both the equations and simplify to obtain the required current through the conductor.

$$ILB = mg$$

$$\Rightarrow I = \frac{mg}{LB} \dots (3)$$

Substitute the values of the known parameters into equation (3) to calculate the required current through the conductor.

$$\begin{aligned} I &= \frac{\frac{40}{1000} \text{ kg} \times 10 \text{ m s}^{-2}}{0.5 \text{ m} \times 0.40 \text{ T}} \\ &= 2 \text{ A} \end{aligned}$$

12. (40)

The magnetic field due to the electron is given by

$$B = \frac{\mu_0 q v \sin 90^\circ}{4\pi r^2}$$

The given data is

$$v = 6.76 \times 10^6 \text{ m s}^{-1}$$

$$r = 0.52 \times 10^{-10} \text{ m}$$

Thus, the magnetic field is

$$B = \frac{10^{-7} \times (1.6 \times 10^{-19}) \times (6.76 \times 10^6)}{0.52 \times 0.52 \times 10^{-20}} = 40 \text{ T}$$